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Degree: B.Tech

Roll no.:.....
Semester: 4

Mid Semester Examination, February 2024

Course Code: CMMTC13

Course title: Applied Linear Algebra

Time: 1 hour 30 minutes

Maximum Marks. 25

Note: Attempt all questions in given order only. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	Let $V = \mathbb{R}^3(R)$ and $S = \{(1, 1, 0), (0, -1, 1), (1, 0, 1)\}$. Find the conditions on a, b , and c such that $(a, b, c) \in L(S)$, where $L(S)$ is the spanning set of S .	2.5	CO1
1b	Let v_1, v_2, v_3 be the vectors of $V(F)$. Show that the set $\{v_1, v_2, v_3\}$ is linearly dependent if the set $\{v_1 + \alpha v_2 + \beta v_3, v_2, v_3\}$ is linearly dependent, where $\alpha, \beta \in F$.	2.5	CO1
2a	Let W be the subspace of $\mathbb{R}^4$ generated by vectors $(1, -2, 5, -3), (2, 3, 1, -4)$, and $(3, 8, -3, -5)$. Find the basis and dimension of W . Extend this basis of W to a basis of $\mathbb{R}^4$.	3	CO1
2b	Check if $W = \{(x, y, z) : z = x + y\}$ is a subspace of $\mathbb{R}^3(R)$ or not. If yes, find the dimension of W .	2	CO1
3a	Let A, B, C be the subspaces of $V = \mathbb{R}^3(R)$ given by $A = \{(a, b, 0) : a, b \in R\}$, $B = \{(0, b, b) : b \in R\}$, and $C = \{(0, b, c) : b, c \in R\}$. Prove that $V = A + B$ and $V = A + C$. Which of these sums is direct? Justify your answer.	2.5	CO1
3b	Let S is a non-empty subset of a vector space $V(F)$. Show that the linear span $L(S)$ is the smallest subspace of V containing S .	2.5	CO1
4a	Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(2, -1) = (1, 1)$, and $T(-1, 1) = (2, 3)$. Find $T(x, y)$ for any $(x, y) \in \mathbb{R}^2$.	2.5	CO2
4b	Let V be the vector space of real polynomials of degree at most 2. Define a linear operator $T : V \rightarrow V$ by $T(x^i) = \sum_{j=0}^i x^j$, $i = 0, 1, 2$. Find the matrix of T^{-1} with respect to basis $\{1, x, x^2\}$ without finding matrix of T .	2.5	CO2
5a	Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by $T(x, y, z) = (x + y - z, x + y + z, y - z)$. Then find the matrix of T with respect to ordered basis $B = \{(0, 1, 0), (0, 0, 1), (1, 0, 0)\}$ of $\mathbb{R}^3$.	2.5	CO2
5b	Find the range, rank, kernel and nullity of the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x - y + 2z, 2x + y, -x - 2y + 2z)$	2.5	CO2